



Interview with Borisov

A video interview with Borisov about finding the comet, in Russian, auto translations are available in English. <https://digit.in/nov19-07>



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the media. The official designation by IAU is 2I/Borisov, where I stands for “interstellar”. The naming convention was established after the discovery of 1I/‘Oumuamua.

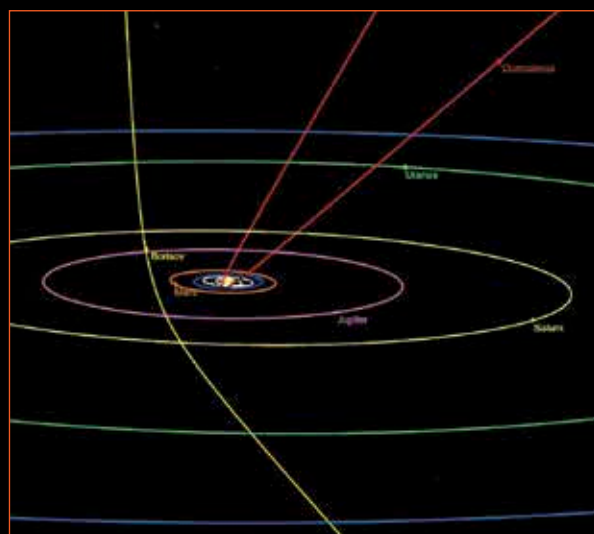
Unlike ‘Oumuamua, 2I/Borisov is not an eccentric body. Except for its high velocity and hyperbolic trajectory, there is really nothing to distinguish it from any other comet within the Solar System. The observations from the various telescopes showed that the object is between 2 to 16 kilometres in diameter. The comet has a slightly reddish hue. The title image for this article shows an image captured by the Gemini North Telescope in two colours, red and blue. The instrument was tracking the comet, so the stars in the background appear as streaks, but actually the stars are stationary while the comet was moving. The image is a composite of four exposures, each 60 seconds long. Andrew Stephens from the Gemini Observatory says, “This image was possible because of Gemini’s ability to rapidly adjust observations and observe objects like this, which have very short windows of visibility. However, we really had to scramble for this one since we got the final details at 3:00 am and were observing it by 4:45!”

It was this image that showed the prominent tail of the comet, and 2I/Borisov was the first interstellar object to show a confirmed outgassing. The outgassing is expected to grow more pronounced as 2I/Borisov approaches closer to the Sun. The escaping gas and particles around the comet is known as a coma. According to estimates, the comet is releasing about 2 kg of dust and 60 kg of water per second. The outgassing is expected to increase significantly in June 2020, when it will be at a distance of between 4 to 5 AU from the Sun. A comet from the Solar System is considered to be “active” at that point. This is when comets grow visible tails, as the volatile material on their surface begins to sublimate. The solid ice on the comet directly turns

to gas. 2I/Borisov is expected to approach within 2 AU of the Sun. One of the most interesting aspects about the comet is how normal it really is. Despite being forged in a star system far away, many millions of years ago, it is still very similar to the comets in our own Solar System. Future observations will allow researchers to characterise the dust, chemical composition and

structure of the object as it passes close to the Sun, allowing researchers to better understand the formation of star systems, including our own. One of the puzzling aspects is exactly why the first interstellar visitor is so different from the second. David Jewitt, lead of the Hubble team says, “Whereas ‘Oumuamua appeared to be a rock, Borisov is really active, more like a normal comet. It’s a puzzle why these two are so different.”

Considering the small size of the planet, astronomers have calculated that there is a ten percent chance that the comet will disintegrate completely. If such a disintegration takes place, there is still expected to be a remnant, which is no longer “active”, such as what has been expected to have happened to ‘Oumuamua. Hubble is scheduled to continue to



The red line indicates the trajectory of ‘Oumuamua, while the yellow line the trajectory of Borisov.

Credit: Tony873004

observe the object during this critical stage.

Just like the months following the discovery of ‘Oumuamua, there are expected to be new studies published on 2I/Borisov. The observation window for the object is much longer than what was available for the first interstellar visitor. The very fact that two interstellar objects were discovered within a span of two years suggests that these categories of objects are much more common than previously believed. Some scientists think that there may be thousands of such visitors at any given point in the Solar System, but they all might be too faint to be detected by the current generation of space telescopes and Earth-based observatories. There are a lot of theoretical implications of this, including recalibration of the potential for interstellar objects to spread life throughout the galaxy. If 2I/Borisov had only entered the Solar System a few years later, it could have been one of the potential targets for ESA’s Comet Interceptor missions, which currently consists of three probes dispatched towards the Oort Cloud. If any future interstellar objects are discovered, then it is possible that one of the Comet Interceptor spacecrafts can closely observe an interstellar visitor. 



Hubble’s observation of 2I/Borisov